SMART AMBULANCE PRIORITY SYSTEM

A DESIGN PROJECT-II (6CS1991 / 6IT1991) REPORT

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In

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Under the Supervision of

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CERTIFICATE

This is to certify that the Design Project - II work entitled "**SMART AMBULANCE PRIORITY SYSTEM**" done by us **CHINTALA VENKATA CHARAN (2023BCSE07AED166) KUNCHAPU PRASAD (2023BCSE07AED167) DASARIRAJU LOKESH (2023BCSE07AED173) KUPPAM DHEEPESH GUPTA (2023BCSE07AED144)** submitted in partial fulfilment of the requirements for the award of the Degree Bachelor of Technology CSE (AIML-B) during academic year 2025-26.

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DECLARATION

I hereby declare that the report of “SMART AMBULANCE PRIORITY SYSTEM” has been prepared for partial fulfilment of the Programme – Bachelor of Technology in Computer Science and Engineering with **Dr. N Sengottaiyan** as guide. I also confirm that the report is the record of my study and prepared as a part of **Design Project -II** work and it is not carried out elsewhere and submitted elsewhere for the award of any degree. Further, I affirm that the variations in the report and the views contained in the report have been discussed and approved by my Mentor as fit.

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ABSTRACT

The problem statement of Overrun of emergency vehicle was very urgent under the heavy Urban traffic conditions. Existing sirens and blinking lights are hardly ineffective for clearing a nearly filled street in the high-density urban environment and subsequently crucial delays in patient care, and road user safety was at risk. Thus, in order to address the above problem, we have implemented a Smart Ambulance Priority System — an IoT-based working prototype that supports vehicle-to-infrastructure communication leveraging on Arduino Uno RF transceiver modules interfacing. The technology enables the ambulance to broadcast a wireless signal, making the traffic signal to green and allow the ambulance the open street to move. LCD is integrated into the system to get a text in display when ambulance or any other emergency vehicle is detected and update of traffic signals. Here, the implemented prototype is demonstrated in a successful first proof-of-concept of signal pre-emption, with the ultimate goal of reducing emergency response time by several minutes and improving overall public safety. Detection of an ambulance is carried out using a distance sensor fixed to the ambulance. Once the ambulance gets within a specific radius of 500 meters, the traffic signal gets automatically changed to green.

LIST OF SYMBOLS & ABBREVIATIONS

Abbreviation	Full Form
EV	Emergency Vehicle
EVP	Emergency Vehicle Preemption
IoT	Internet of Things
ITS	Intelligent Transportation Systems
V2I	Vehicle-to-Infrastructure (Communication)
V2V	Vehicle-to-Vehicle (Communication)
RF	Radio Frequency
GPS	Global Positioning System
TMC	Traffic Management Center
IR	Infrared (Sensor)
PIC	Peripheral Interface Controller (Microcontroller type)
LoRa	Long Range (Communication Protocol)

CONTENTS

1. INTRODUCTION

The world is urbanizing at a faster rate than ever before, and good emergency services are vital for the safety of cities and in saving lives. One of the severest problems that today's cities have to address is traffic congestion, leading to significant delays for emergency vehicles (EVs) such as ambulances. Rapid response matters so much for medical emergencies — one study found that just a one-minute delay decreases the survival odds of patients with cardiac arrest by 7% to 10%.
The traditional traffic management system is designed for regulating routine traffic, and it is not tailored to emergency control.

To fill this gap, we propose a Smart Ambulance Priority System as an automated solution for real-time priority assistance to ambulance vehicles through interaction with traffic infrastructure. The project uses Vehicle-to-Infrastructure (V2I) as a communication technique, where the ambulance transmits its priority signal using Arduino Uno and RF transceiver modules for wireless transmission. The traffic light automatically turns green, allowing a clear and safe path for the emergency vehicle. Conventional traffic systems use control through human intervention or siren usage, but they can be affected by noise. This project uses sensors for better accuracy.

By doing so, our system demonstrates the effectiveness of automatic signal preemption in significantly reducing emergency response time, ultimately enhancing public safety and contributing to the evolution of smarter, safer, and more responsive urban cities of the future.

2. LITERATURE REVIEW

The review below details the state-of-the-art research and commercial offerings in relation to Emergency Vehicle Preemption (EVP) and intelligent traffic management, presenting an overview of current technological capabilities and the specific gaps addressed by our prototype.

Paper 1: A Strategy for Emergency Vehicle Preemption and Route Selection

Abstract:

The idea behind EVP is that EV that takes the right-of-way to the fire via a green wave en-route. However, most existing formulations only consider how to minimize EV's travel time and entirely neglect the impact on normal traffic. Thus, this paper formulates a joint optimal path selection and EV preemption strategy. Instead of implementing the emergency preemption just when the vehicle starts to move, the preempting guards all the intersections in the path to the fire short before EV reaches them by choosing which corridor to preempt and when to start preempting call. The proposed path-aware scheduling strategy is tested via microscopic simulation, demonstrating the significant decrease of EV's travel time and the minor impact on normal traffic.

Accomplishments and Limitations:

- **Priority for EVs:** Modern systems like SCATS, SCOOT, RHODES, and ACS Lite are intended to eliminate the need for stops, helping emergency vehicles move more efficiently.
- **Green Wave Creation:** EVP systems temporarily adjust standard signal patterns to form a continuous "green wave" for EVs. This allows them to travel smoothly and quickly along their route while minimizing potential traffic conflicts.
- **Existing Systems:** Several preemption systems are already in operation, including OPTICOM, BLISS, STROBECOM, and MIRT.

Scope:

The approach was evaluated in a microscopic simulation (VISSIM) envi-ronment, based on a real arterial network. The results indicated that the travel time of the EV for each scenario all had been reduced significantly from 15.8% to 48.8%, and meanwhile causing minimum influence on the general traffic. In general, EVs on the mainstream approach suffered less lost time and enjoyed improved progression because of the longer green coverage. The joint path selection and dynamic preemption method and its evaluation can serve as a practical tool for agencies and practitioners to evaluate the potential efficiency of EVP on arterials that are either already built or under planning.

Paper 2: Smart Ambulance Using IOT

Abstract:

Traffic jams is one of the biggest major issue in India because of which there is a lot of damage to the ambulance services, waiting time may cause life loss for a patient and such incidents are increasing every day. In each of the above, there aint no patient satisfaction cuz he ain't home. So we invented the idea of giving him what he was asking for. In the smart ambulance, several sensors like Heart rate sensor, blood pressure, ECG and so on as mentioned above sensing sensors will analyze the condition of all these vital parameters and status of these sensors will be stored along with transmitted in hospital's database by wireless via mobile GPRS message at a time signals given by using cloud. As soon as the critical condition that is needed by our system get recorded, then we will record it to calculate it, and at a time (we will send such type of cardiac attack or diseases to hos-pital by using GPRS and also accomplish our primary requirement, because we de-signed our sys-tem in this way). To came to our patient life-saving we drove the hospital as a system is byyor we went to green if A signal between them becomes more 30m and identifies another system green, by GPRS symptoms communicate in cloud. Our sys-tem isacontrol entity orambulance on its side, controlsthe trafficlights' anditautomaticallyswitches to the hospital.

Accomplishments and Limitations:

- Creates a **"Green Corridor":** when the ambulance approaches the 100m range, the system automatically takes control of traffic signals and converts them from red to green to guarantee a special, uncluttered path to the destination.
- Showed real-time adaptability in traffic signal management.

Scope:

Less Traffic Impact Recovery Mention: That is, even the abstract says that the system "provides for a quick response," the impact of reducing the negative effect on general traffic flow, signal coordination, and how shortly the normal process of traffic works and strings are restored after the preemption ends is not considered comprehensively.

3. IDENTIFICATION OF THE PROBLEM / SCOPE OF THE PROJECT

a) Problem Statement

Ambulances get stuck in traffic, which can make all the difference when it comes to patient care and public safety. Depending on sirens and visual warnings tends to be in vain when driving through heavy urban traffic, where congestion drowns out the sound signals and prevents drivers from yielding properly. This inefficiency, particularly at signalized intersections, poses a direct danger to public health. According to the American Heart Association, a cardiac arrest patient's chance of survival can drop by **7 to 10%** for every minute that treatment is delayed. Hence, there is an urgent need for an automated system capable of securing the right-of-way for emergency responders in a reliable and safe manner.

b) Scope of the Project

The scope specifies the limits and objectives of this prototype. It is developed as a **proof of concept**, focusing on the fundamental aspects of communication and control logic rather than large-scale implementation.

In-Scope Functionalities (What We Are Doing):

- **V2I Communication Proof-of-Concept:** Establishing a reliable, one-way RF (Radio Frequency) wireless link between a single Ambulance Unit (Transmitter) and a single Traffic Signal Unit (Receiver) to demonstrate the core principle of Vehicle-to-Infrastructure communication.
- **Unique EV Identification:** Using the NRF24L01 transceiver module (or 433MHz RF with unique encoding) to securely recognize the emergency vehicle and block any unauthorized preemption requests.
- **Automated Preemption Logic:** Designing and developing state machine logic that governs the signal override sequence — transitioning from Yellow (Caution) → All-Red (Safety Pause) → Green (Clearance for Emergency Vehicle).
- **Proactive Information System:** Incorporating Ultrasonic Sensors to gather real-time traffic density data at intersections and transmitting it to an operator or dispatcher interface for better traffic management awareness.
- **Cost-Effective Prototype:** Creating a low-cost proof-of-concept that highlights the complexity of V2I technology using affordable and easily available hardware components such as the Arduino Uno and RF modules.

4. PROPOSED IDEA

Our proposed solution is the Smart Ambulance Priority System — an intelligent, automated, and wireless control framework designed to reduce the critical delays faced by emergency vehicles at static urban traffic intersections. The prototype enables direct communication between the ambulance and the traffic control system, ensuring a safe, quick, and reliable right-of-way during emergencies.

a) System Overview

The foundation of this system is the Vehicle-to-Infrastructure (V2I) communication concept, which involves two major components:

- Ambulance Unit (Transmitter)
- Traffic Signal Unit (Receiver)

Wireless Communication Backbone: The system employs RF (Radio Frequency) technology for dependable, short-range wireless data transmission. The ambulance sends a secure, encoded priority signal directly to the nearby traffic light controller.

Automated Response: Upon receiving and validating this signal, the Traffic Signal Unit overrides the normal light sequence to activate an emergency protocol, ensuring that the ambulance receives a green signal while other directions are safely stopped.

Clear Corridor Creation: This process immediately clears the intersection, providing an unobstructed route for the ambulance to move through without delay.

b) Key System Functions

Unique EV Identification:

The Ambulance Unit is fitted with an NRF24L01 transceiver module (or an equivalent RF module with unique encoding) to ensure that only authenticated emergency vehicles can initiate a preemption request, thus preventing unauthorized overrides.

State Machine Control:

The Traffic Signal Unit runs on finite state machine logic, executing a carefully designed safety sequence upon receiving a priority signal — typically:

Yellow (Caution) → All-Red (Safety Pause) → Green (Clearance for Emergency Vehicle).

Proactive Data Gathering:

Ultrasonic Sensors are deployed at intersections to monitor real-time traffic density. This data can be transmitted to a central operator or dispatcher, helping them make proactive routing decisions to prevent the ambulance from getting stuck in traffic.

c) Benefits of the Proposed Solution

Reduced Response Time:

By automatically preempting red lights, the system minimizes waiting time, thus significantly cutting down critical emergency response durations.

Enhanced Safety:

Automating signal overrides minimizes the risk of collisions caused by emergency vehicles running red lights and reduces confusion among other drivers.

Cost-Effectiveness:

This prototype demonstrates advanced signal preemption and V2I communication principles using affordable hardware such as Arduino Uno and standard RF modules, making it a highly accessible and scalable proof-of-concept.

5. OBJECTIVES OF THE IDEA

The development of the **Smart Ambulance Priority System prototype** is guided by a set of clearly defined objectives aimed at ensuring a technically robust, innovative, and impactful demonstration of its potential within the scope of a design project.

5.1. Functional Prototype Development:

To design and construct a working prototype of a smart traffic signal preemption system using readily available, cost-effective hardware components.

5.2. Reliable Wireless Communication:

To develop and validate a stable RF wireless communication link between the Ambulance Unit (Transmitter) and the Traffic Signal Unit (Receiver), ensuring consistent data transmission upto radius of 500 meters.

5.3. Automated Priority Logic:

To implement the core State Machine Logic that enables automatic traffic light control. The logic should execute a safe transition sequence — *Yellow (Caution) → All-Red (Safety Pause) → Green (Clearance for EV)* — and safely revert the signal back to the normal traffic cycle.

5.4. Authentic EV Identification:

To employ the NRF24L01 transceiver module (or a similar RF module with encoded ID) to ensure unique identification and authentication of emergency vehicles, thereby preventing unauthorized preemption requests.

5.5. Enhanced Situational Awareness:

To integrate Ultrasonic Sensors for precise vehicle density detection and transmit this real-time intersection data to an operator or dispatcher interface, enabling proactive route planning.

6. METHODOLOGY

The Smart Ambulance Priority System was developed using a structured and iterative methodology, grounded in the core principles of Design Thinking to ensure that the system is both technically reliable and deeply aligned with user needs. This approach enables the project to directly tackle the urgent problem of emergency vehicle delays in urban traffic.

6.1 The first stage involved developing an empathetic and thorough understanding of the challenges faced by key users:

- **Primary Customers (Ambulance / Emergency Services):**
  Their primary need is uninterrupted and safe passage through intersections. They require a reliable, simple-to-use system that drastically reduces time lost at red lights.
- **Secondary Customers (Patients):**
  Their essential need is speed and reduced response time, which directly impacts survival outcomes.
- **Users (Traffic Managers / Dispatchers):**
  They require situational awareness, meaning real-time knowledge of intersection congestion to make proactive routing decisions.

6.2 Each identified need was directly converted into a technical function within the prototype:

Identified Need	Project Translation (Product Function)
Unimpeded Passage	Automated Preemption Logic (Traffic Signal Unit overrides normal cycle)
Reliability & Authentication	RF Transceiver Module with encoded priority signal to prevent false triggers
Situational Awareness	Ultrasonic Sensor Module integrated with operator feedback system
Ease of Use	Simple Activation Button on the Ambulance Unit to initiate the signal sequence

6.3 Specification of the Product / Process / Project

This stage established the system's technical specifications and operational framework:

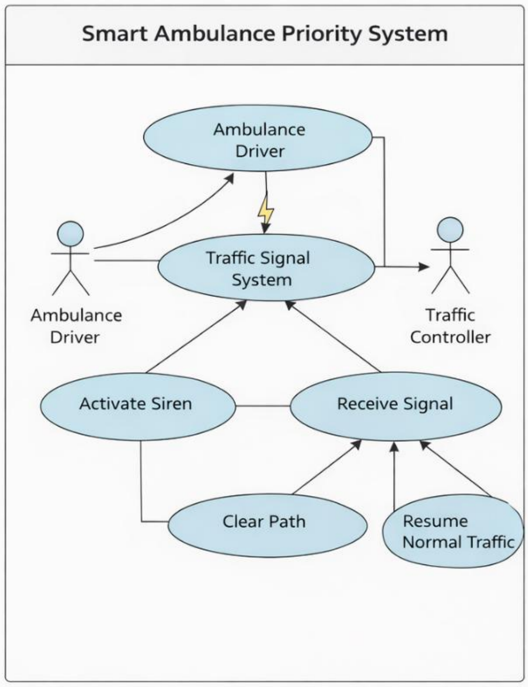
- **System Architecture:**
  Defined as a two-part embedded system implementing Vehicle-to-Infrastructure (V2I) communication.
- **Transmitter Unit:**
  Built using Arduino Uno with a 433MHz RF Transmitter module.

- **Receiver Unit:**
  Constructed using another Arduino Uno paired with a 433MHz RF Receiver.
- **Logic Specification:**
  Designed as a finite state machine, incorporating a crucial all-red safety pause between conflicting traffic phases to ensure intersection safety.
- **Sensing Specification:**
  Replaced traditional IR sensors (common in literature but unreliable in fog or rain) with Ultrasonic Sensors for more accurate, real-time vehicle detection.

6.4 This phase involved hardware integration and software development based on the established specifications:

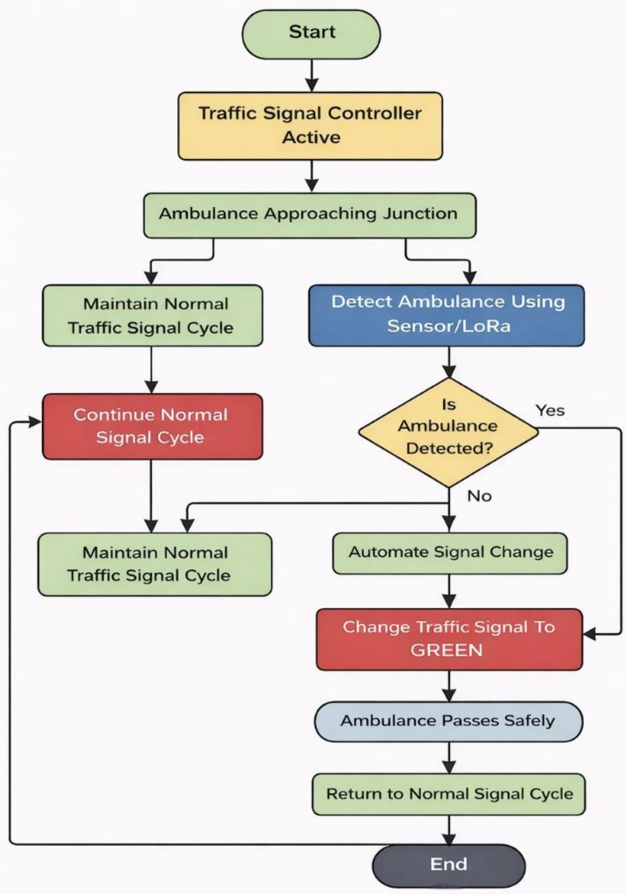
- **Hardware Interfacing:**
  Circuits were designed to connect RF modules, LEDs, and sensors to the Arduino boards, forming the physical communication and control network.
- **Software Development (Transmitter):**
  Event-driven code was developed for the Ambulance Unit, where pressing a button triggers a sequence that encodes and transmits a unique identifier to the receiver.
- **Software Development (Receiver):**
  The Traffic Signal Unit continuously listens for incoming signals, verifies the authentication code, and executes the predefined Light Control Logic to manage the signal override.
- **Flow Implementation:**
  The software manages the entire operational cycle — from receiving the emergency signal to executing the light transition protocol (Yellow → All-Red → Green for Ambulance) and finally returning to normal operation once the ambulance has passed.

6.5 Flow Chart



Use Case Diagram

Activity Diagram of Smart Ambulance Priority System



7. SIMULATION

To ensure the functionality of the Smart Ambulance Priority System before actual implementation, the entire circuit was designed and simulated using Autodesk Tinkercad Circuits. This simulation helped in the verification of the circuit connections, transitions, and emergency override.

7.1 Simulation Overview

This Simulation consists of two Major units:

1. Ambulance Unit (Transmitter):

- Arduino Uno microcontroller
- Activation switch (symbolizes ambulance siren ON)
- RF transmitter module
- Ultrasonic sensor (for vehicle detection)

2. Traffic Signal Unit (Reciever):

- Arduino Uno microcontroller
- RF receiver module
- Red, Yellow, and Green LEDs
- LCD display (for status monitoring)
- Ultrasonic sensor

7.2 Working of Simulation

The simulation uses a state machine logic that involves two modes:

- Normal Mode

  Traffic lights have a set cycle that involves:

  Red → Green → Yellow

  This ensures smooth flow under normal conditions.

- Emergency Mode

  When the ambulance is detected within range (~500 meters):

  Red → Yellow → Green

  The GREEN light remains green for 10 seconds to enable the safe passage of the ambulance.

- Return to Normal Mode

  After the ambulance crosses:

  Green → Yellow → Red

  The system automatically resumes normal traffic flow.

7.3 Simulation Output

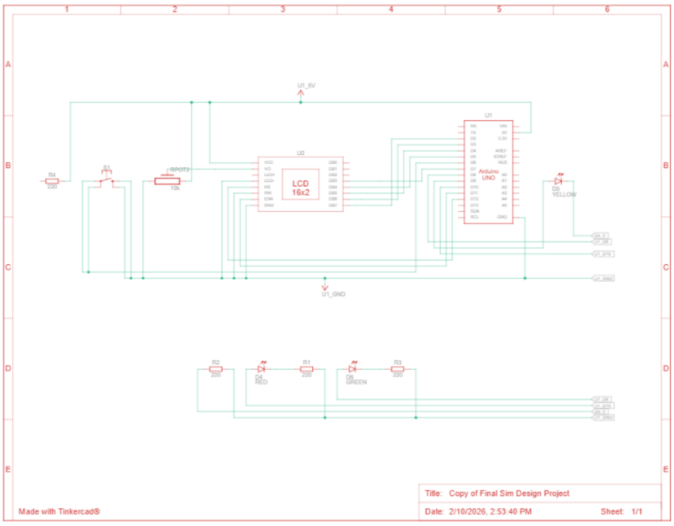


Fig 1 Circuit Diagram

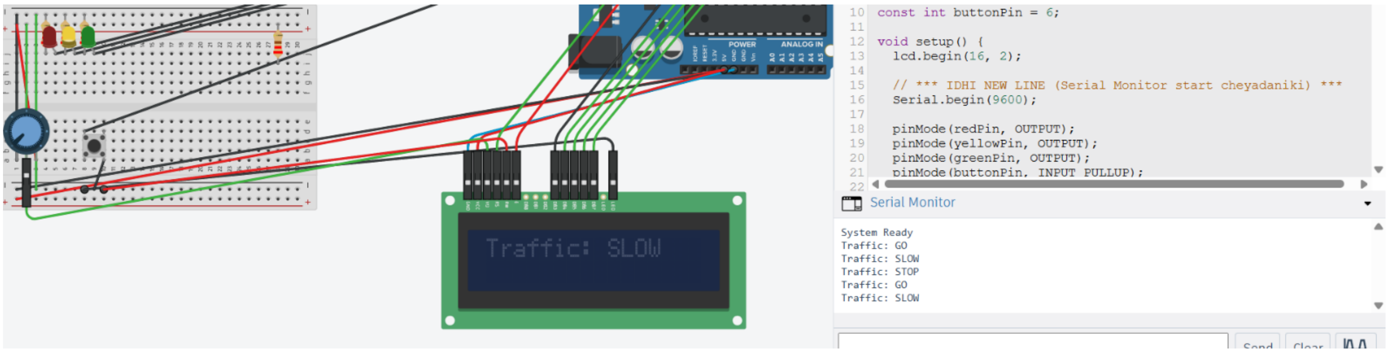


Fig 2 Traffic Light Simulation

Fig 3 Emergency Override

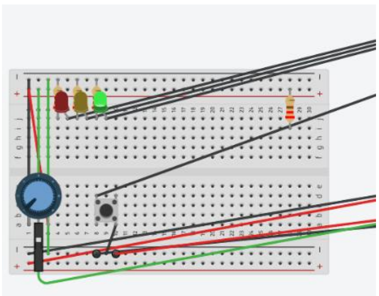


Fig 4 LCD output

8. REQUIREMENTS

This section outlines the essential hardware, software, and tools used in the development and prototyping of the Smart Ambulance Priority System.

a) Hardware Requirements:

Component Category	Component	Function in Project
Microcontrollers (CPUs)	**Arduino Uno**	This serves as the CPU of both the Ambulance (Transmitter Part) and Traffic Signal (Receiver part).
Communication	**433MHz RF Transmitter Module**	Installed in the Ambulance Unit to broadcast the encoded priority signal wirelessly.
	433MHz RF Receiver Module	Installed in the Traffic Signal Unit to continuously monitor and receive priority signals within approximately 500 meters.
Input	**Activation Button**	Triggered by emergency personnel to initiate the RF transmission sequence.
	Ultrasonic Sensor (e.g., HC-SR04)	Detects vehicle presence and density at the intersection an enhancement over IR-based sensors.
Output	**LEDs (Red, Yellow, Green)**	Simulate traffic light phases and visualize the signal override sequence.
Support Components	**Power Supply (9V or USB), Breadboards, Jumper Wires, Resistors, LCD Display**	Provide power and enable safe circuit construction for the prototype and LCD will display the EMERGENCY text when ambulance detected.

b) Software Requirements:

- **Integrated Development Environment (IDE):** Arduino IDE — used for writing, compiling, and uploading the control code to the microcontrollers.
- **Programming Language:** Embedded C/C++ used to write Arduino sketches implementing logic, communication, and signal transitions.

c) Technologies and Tools:

- **Core Technology:** Embedded Systems — integration of software algorithms with electronic hardware components.
- **Communication Protocol:** Radio Frequency (RF) — ensures secure and reliable wireless data transmission, including signal encoding and verification.
- **Programming Principle:** Event-Driven Programming — the system responds to specific input triggers such as button press or signal reception.
- **Logic Model:** State Machine Logic — governs traffic light behavior as a finite-state automaton, managing transitions between Normal Operation and Emergency Override modes.
- **CircuiDesign Tools:** Fritzing (or an equivalent circuit design tool) used for creating and verifying schematic diagrams before implementation.

9. NOVELTY OF THE PROJECT

Proactive Dispatch Decision Support

Unlike most systems that react after an EV is within close proximity to the intersection, our system also features real-time delivery of traffic junction status information to the operator. This vital functionality allows dispatchers to reroute a vehicle when congestion is detected before it even starts the journey, saving critical response time. This directly resolves the limitation seen in studies where the EV must first become stuck before action is taken.

Improved and Robust Vehicle Detection

We exploit Ultrasonic sensors to provide highly accurate vehicle detection at the intersection approaches. This marks a clear technical advancement over earlier methods that used IR sensors, which are prone to sunlight interference and thus unreliable in outdoor conditions.

Secure and Authenticated EV Identification

We use the NRF24L01 transceiver module (or RF with encoded ID) to establish a secure communication channel with a unique identifier. This mechanism specifically differentiates emergency vehicles from others and grants them priority — addressing the lack of clear identification protocols seen in prior systems.

Concurrent Real-World Awareness and Priority Control

Our system delivers the advantages of sophisticated commercial systems (like *Glance*, which clears traffic ahead of arrival) while adding the benefit of operator awareness. This dual capability allows the operator to monitor every emergency intersection's status before authorizing an emergency vehicle to begin its route.

Low-Cost, Easy-To-Build Prototype

We demonstrate high-level Emergency Vehicle Preemption (EVP) concepts using inexpensive and widely available Arduino microcontrollers and RF modules. This approach makes the core technology accessible for education and rapid prototyping, avoiding the major infrastructure costs of high-end GPS or cellular-based systems.

10. CONCLUSION

Finally, through the development and deployment of this project, a working prototype of the Smart Ambulance Priority System has been designed and implemented — presenting an elegant, affordable, and highly relevant solution to the persistent issue of emergency vehicle delays in cities. We validate the concept that automated, event-driven priority, powered by Vehicle-to-Infrastructure (V2I) communication through embedded systems (Arduino and RF modules), can significantly reduce emergency response times.

The system directly tackles the life-threatening impact of traffic congestion by providing a reliable mechanism for traffic signals to automatically execute an emergency override protocol.

The originality of the system is based on:

- **Increased Reliability:** Uses Ultrasonic sensors for accurate vehicle detection.
- **Secure Identification:** Employs the NRF24L01 transceiver for unique, authenticated communication, ensuring that only genuine emergency vehicles receive signal priority.
- **Proactive Planning:** Provides operators with real-time intersection data, enabling preemptive route optimization before the journey begins.

This project represents a pivotal step toward smarter, more human-centered traffic infrastructure. By successfully creating and demonstrating this functional prototype, we contribute to the broader goal of building safer cities and improving outcomes for patients in medical emergencies.

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